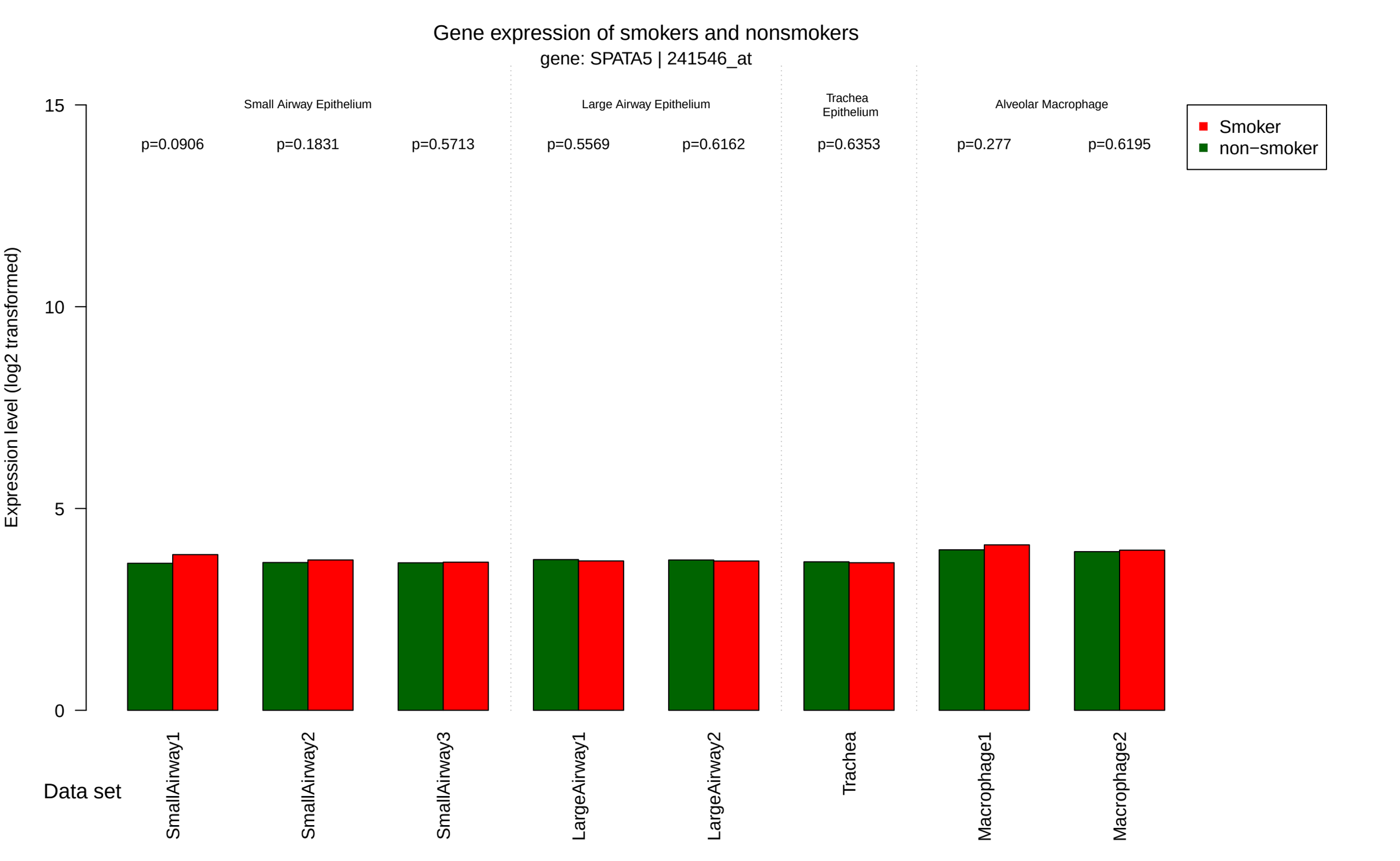
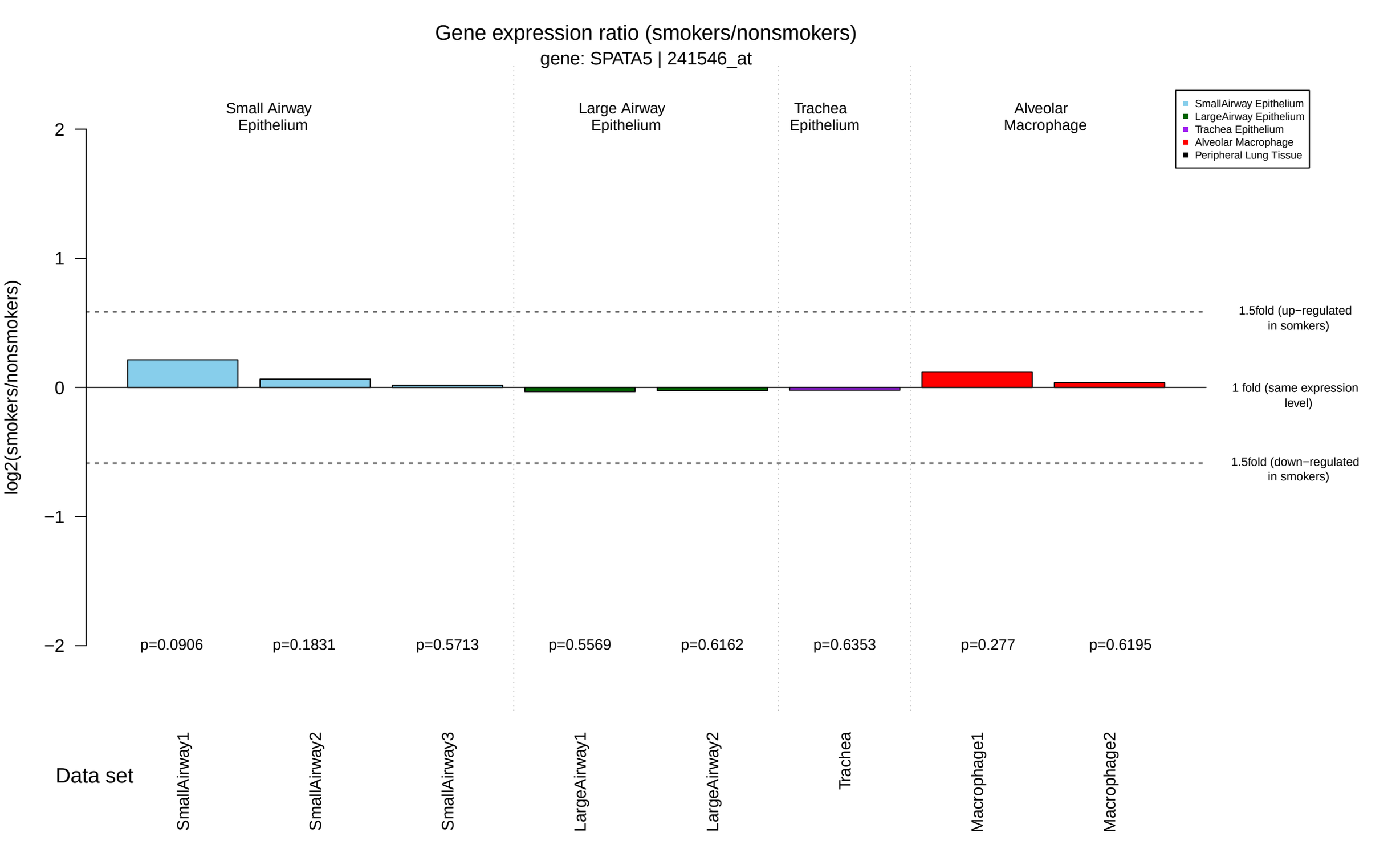
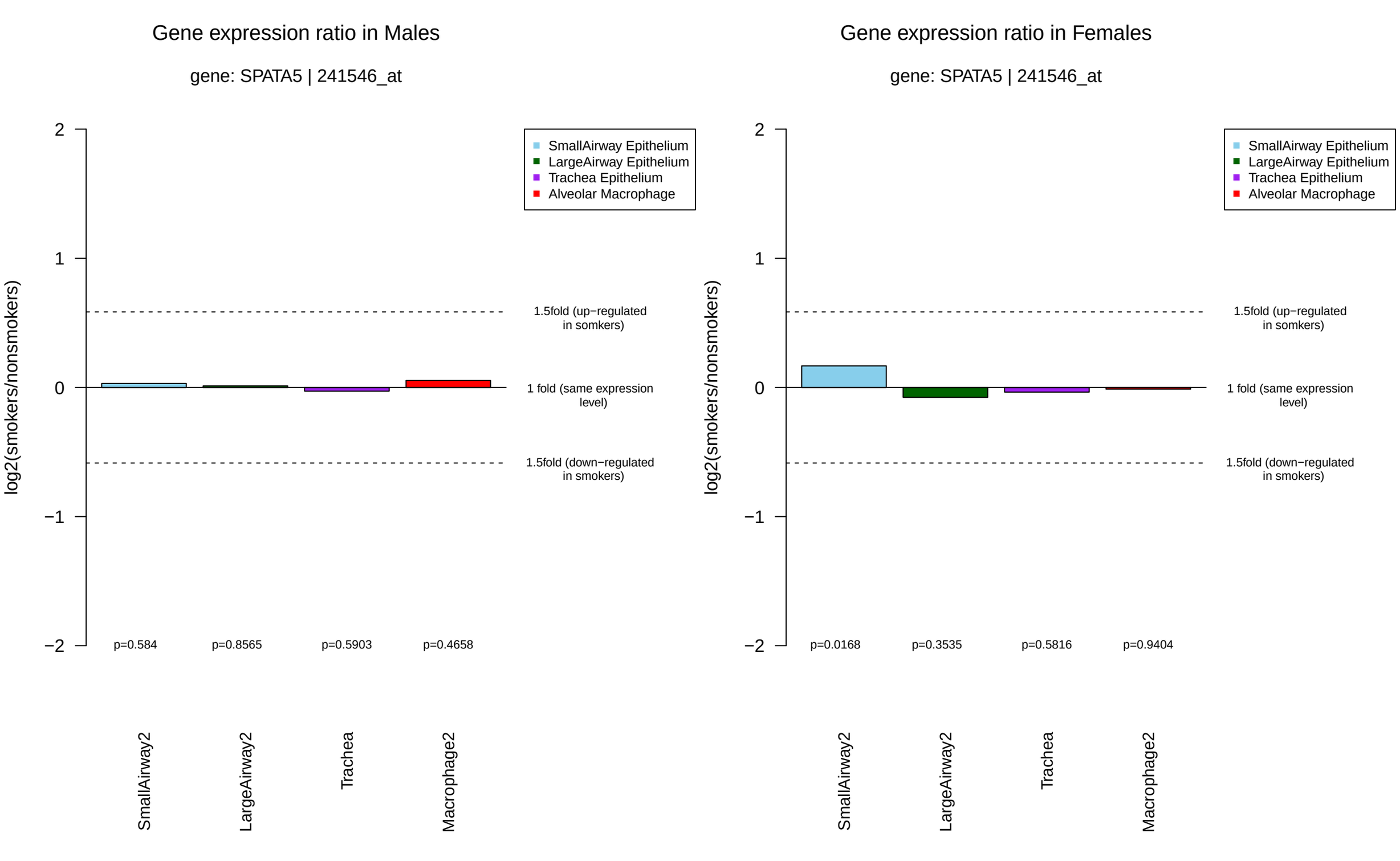
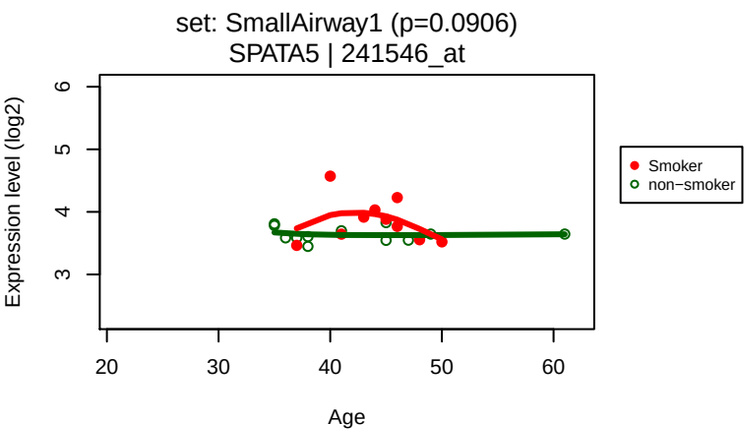
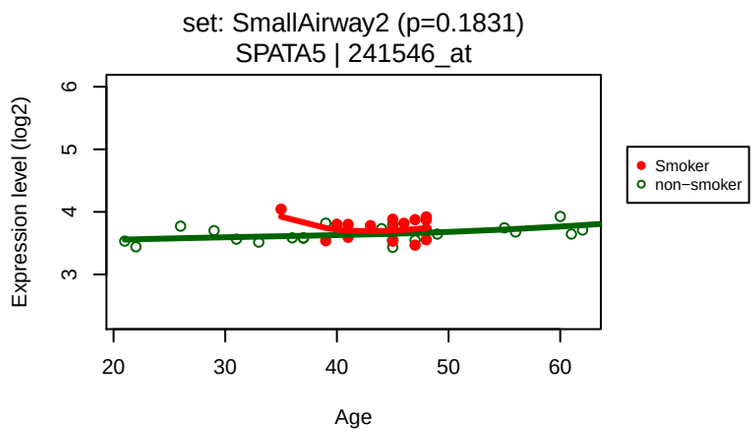
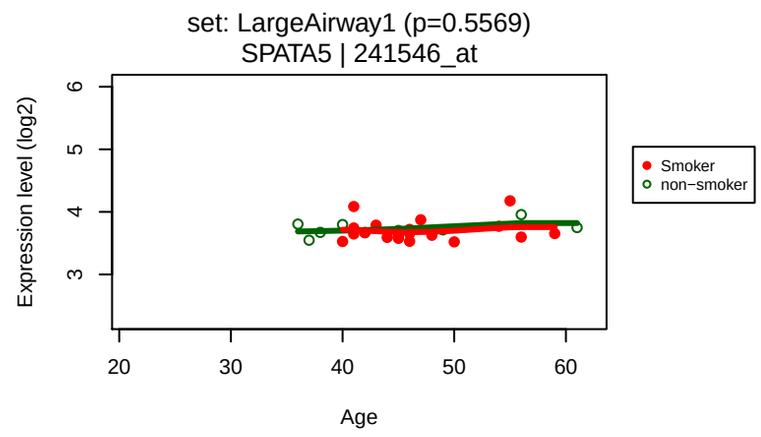

Gene expression of smokers and nonsmokers
gene: SPATA5 | 241546_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0906
p=0.1831
p=0.5713
p=0.5569
p=0.6162
p=0.6353
p=0.277
p=0.6195
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

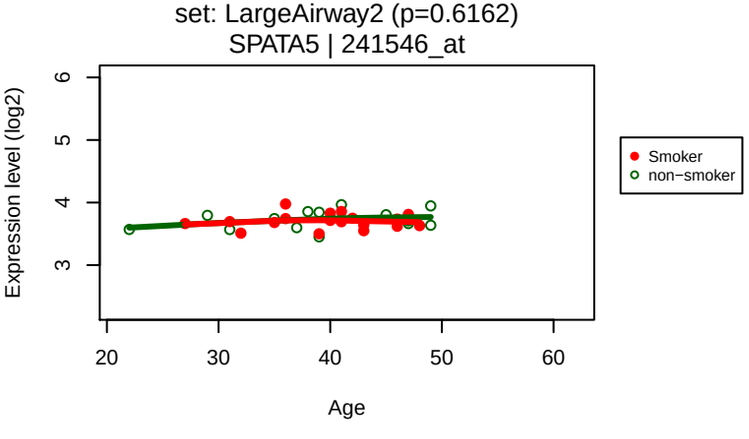

Gene expression ratio (smokers/nonsmokers)
gene: SPATA5 | 241546_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0906
p=0.1831
p=0.5713
p=0.5569
p=0.6162
p=0.6353
p=0.277
p=0.6195
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

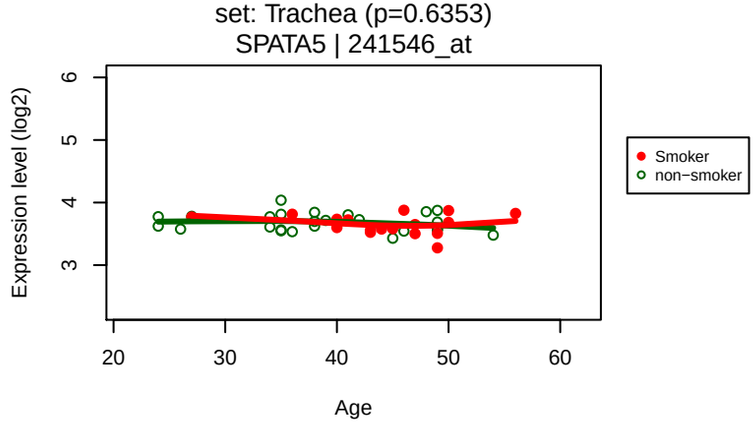

Gene expression ratio in Males
gene: SPATA5 | 241546_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.584
p=0.8565
p=0.5903
p=0.4658
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPATA5 | 241546_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0168
p=0.3535
p=0.5816
p=0.9404
SmallAirway2
LargeAirway2
Trachea
Macrophage2

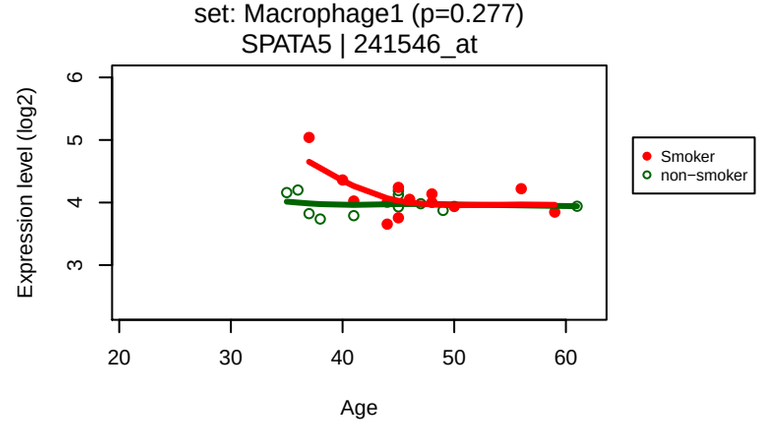

set: SmallAirway1 (p=0.0906)
SPATA5 | 241546_at
Expression level (log2)
Age
Smoker
non−smoker

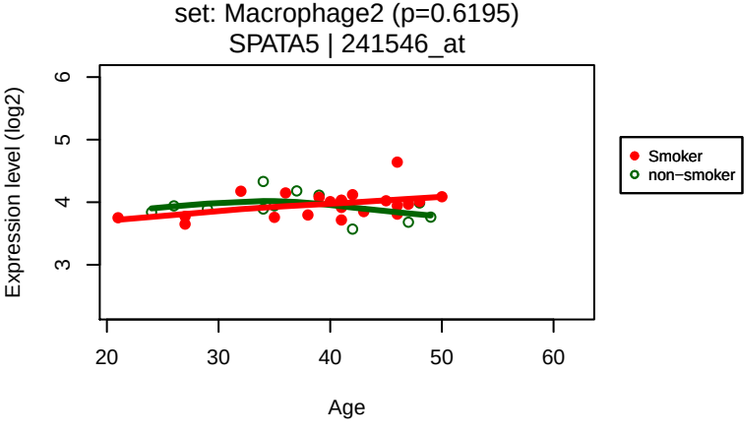

set: SmallAirway2 (p=0.1831)
SPATA5 | 241546_at
Expression level (log2)
Age
Smoker
non−smoker

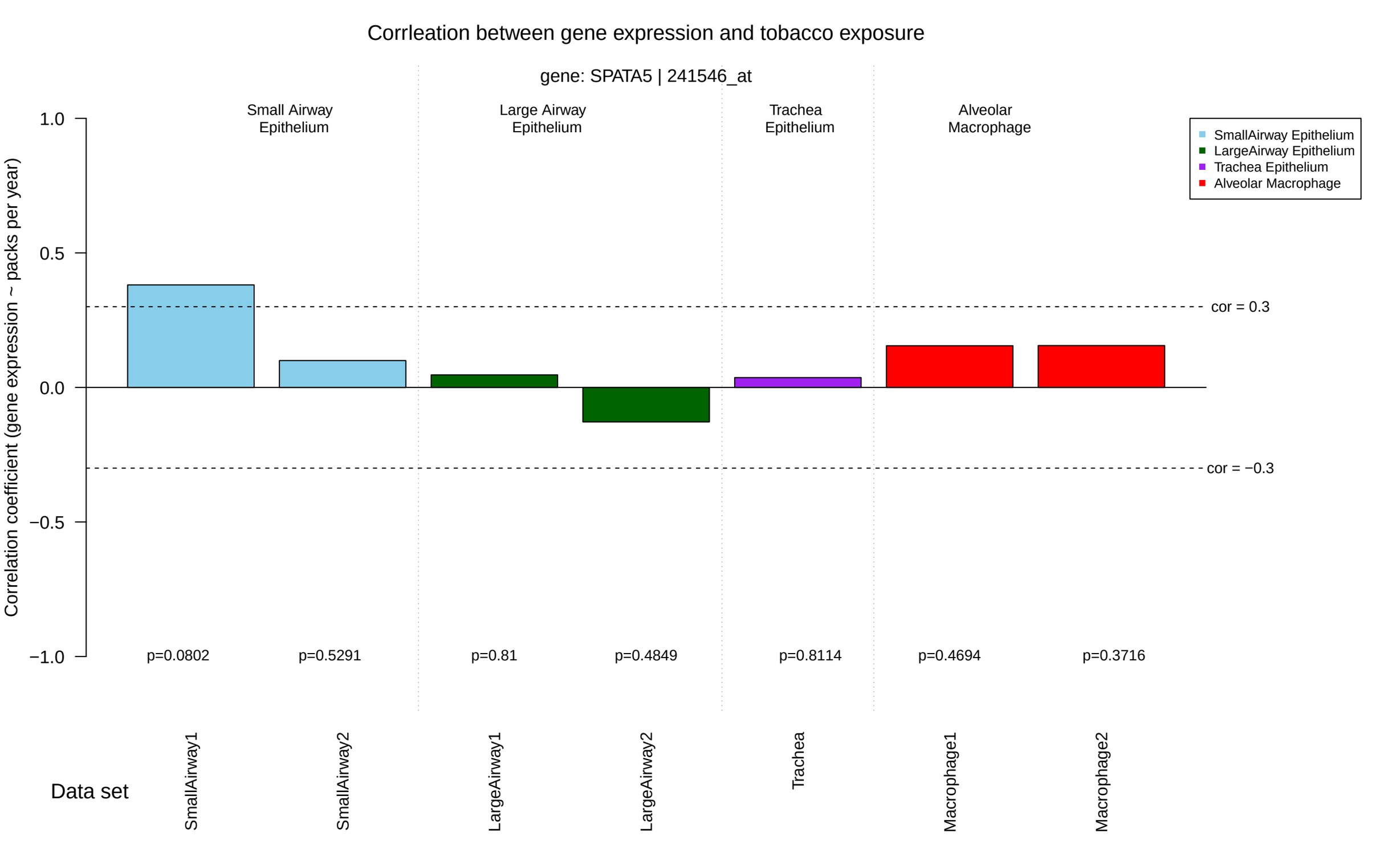

set: LargeAirway1 (p=0.5569)
SPATA5 | 241546_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6162)
SPATA5 | 241546_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6353)
SPATA5 | 241546_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.277)
SPATA5 | 241546_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6195)
SPATA5 | 241546_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPATA5 | 241546_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0802
p=0.5291
p=0.81
p=0.4849
p=0.8114
p=0.4694
p=0.3716
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2